FIGURE MATRIX

Rows, Columns, Cycles and the Missing Figure

OAVS TGT | Reasoning (Non-Verbal) | Topic 17: eighth non-verbal topic

Syllabus line (Reasoning, 30 Marks): “The Non-Verbal Reasoning topics like Figure Formation, Dice, Triangle, Rule Detection, Images, Completion of Pattern, Mirror Images, **Figure Matrix** and Paper Folding, etc.”

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). Only two figure-matrix questions appear in these papers (English 2019 Q53 and English 2022 Q47). Both are solved with their original figures. The practice questions were generated by program so that each has exactly one option that fits.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	The 2 PYQs, the rules they used, priority list
2	Foundation	7 matrix rules and the row-column-diagonal method
3	Solved PYQs	English 2019 Q53 and English 2022 Q47 with original figures
4	Shortcuts, Traps and Scope	Speed methods and mistakes
5	Practice Set	6 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

Two questions, two different rules

Paper	Q.	Matrix	Rule used	Ans
English 2019	53	3 × 3, shapes with layers	Row progression: each step adds a layer (single shape, then doubled shape, then doubled shape inside circles)	1
English 2022	47	3 × 3, shapes inside shapes	Cyclic: circle, triangle and square cycle through rows and columns; inner shape is the next in the cycle	A
Mathematics 2024	–		none	
Science 2024	–		none	
Social Studies 2023	–		none	

PYQ INSIGHT: Both PYQs are 3 × 3 matrices with the question mark in the **bottom-right cell** and four options. Each option set is built from near-misses of the correct figure: a missing layer, a wrong shape or an extra part. In both PYQs the answer comes from reading **one row**, not the columns. The figure-series questions in the other papers (a row of figures with a question mark) are a different type (Rule Detection).

Rule type	PYQs	Priority
Row progression (a layer or part is added step by step)	1	★★★ Must
Cyclic or Latin arrangement (each shape once per row and column)	1	★★★ Must
Overlay and common-part removal across a row or column	0	★★ Expected
Rotation by a fixed angle	0	★★ Expected
Counting (dots, lines, sides add or subtract)	0	★ Possible

CHAPTER 2

Foundation

Seven matrix rules and a fixed method

2.1 What a figure matrix is

A 3 × 3 grid of figures follows one rule that works on every complete row (or every complete column). The last cell is missing. Find the rule from the complete lines, then apply it to the incomplete one.

2.2 The seven rules

Rule	What happens	Example
1. Progression	A part is added (or removed) at each step along the line	Single shape, doubled shape, doubled shape in circles (Eng 2019 Q53)
2. Overlay (addition)	Third figure = first and second drawn together	Circle + plus gives circle with plus
3. Common removal (subtraction)	Parts common to the first two vanish; the rest form the third	Circle+triangle and square+triangle give circle+square
4. Cyclic / Latin	Each shape (or position) appears once in every row and column	Eng 2022 Q47
5. Rotation	Each figure is the previous one turned by a fixed angle	0°, 90°, 180° clockwise along a row
6. Counting	Number of dots, lines or sides adds, subtracts or increases by a fixed step	1 + 2 = 3 dots
7. Position or shading change	A dot or shading moves or fills by a fixed step	Shading moves one cell clockwise

2.3 The method (do not skip the order)

Step	Do this
1	List the features of each figure: outer shape, inner shape, lines, dots, shading, size, turn.
2	Test the **rows**: which features stay, which change, and how (add, overlay, rotate, count).
3	If the rows give no rule, test the **columns**; then the diagonals; then check for a cyclic or Latin pattern.
4	Check the rule on **every** complete line, not just one, and predict the missing figure before looking at the options.
5	Eliminate options with a wrong shape, a missing or extra part, or the wrong count.

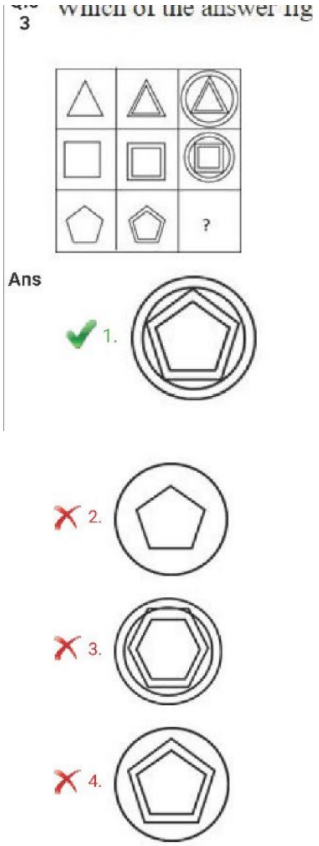
SHORTCUT: Predict, then match: draw the answer in your head from the rule, then pick the option that looks like it. Near-miss options are designed to tempt you if you compare first.

NOTE: Rows 1 and 2 can be decoration. In Eng 2022 Q47 the lines inside the figures of rows 1 and 2 are not part of the rule for row 3; the nested shapes of row 3 and the cycle of outer shapes decide the answer.

CHAPTER 3

Solved PYQs

Both questions with original figures



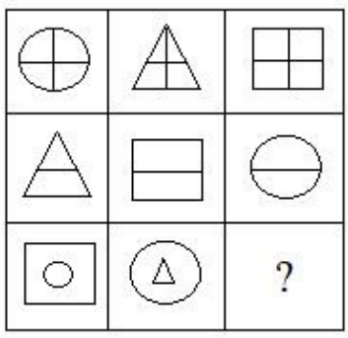
English 2019 Q53

Which of the answer figures completes the given figure matrix?

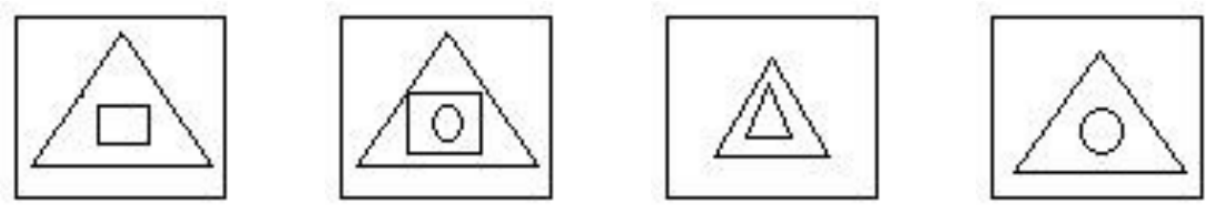
1. Row 1: a triangle, then the triangle drawn twice (one inside the other), then the doubled triangle inside **two circles**.
2. Row 2 does the same with a square: single square, doubled square, doubled square inside two circles.
3. So the rule is: column 1 is the single shape; column 2 adds an inner copy; column 3 adds two circles around the column 2 figure.
4. Row 3 starts with a pentagon, so the missing figure is a **doubled pentagon inside two circles**.
5. Option 2 (single pentagon, one circle) and option 4 (doubled pentagon, only one circle) are missing layers. Option 3 is a hexagon, the wrong shape. Only option 1 fits.
Answer: Option 1

English 2022 Q47

Select the figure from the options that can replace the question mark (?) and complete the pattern.



1. Outer shapes: row 1 is circle, triangle, square; row 2 is triangle, square, circle; row 3 starts with square, circle. Each shape appears once in every row and column, so the missing outer shape is a **triangle**.
2. Row 3 has a shape inside each outer shape: a circle inside the square, a triangle inside the circle. Each inner shape is the outer shape of the next cell (circle, triangle, then the first cell's outer shape, square).
3. So the missing figure is a **triangle with a small square inside**.
4. B has two inner shapes (square and circle); C has a triangle inside; D has a circle inside. Only A is a triangle with a square.
Answer: Option A



A ✓ **B** **C** **D**

NOTE: The scan of Eng 2019 Q53 is small, so the layer counts are read from the figure: column 3 shows the doubled shape inside two circles in rows 1 and 2, and option 1 is the only option with the same layers for the pentagon. The official key is Option 1.

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

- **Row first, column second:** both PYQs were solved from a single row.
- **Count layers:** number of outlines, circles and dots decides most matrix questions.
- **Third = first + second?** Test overlay and common-removal in the first row at once; they are the most common.
- **Shape cycle:** if each of three shapes appears once in every row and column, look for a cyclic rule.
- **Fixed turn:** if figures look the same but face different ways, test a constant rotation.
- **Time:** 40 seconds per question.

Trap	How to avoid it
Using one row only	Check the rule on every complete row (or column) before using it.
Picking the busiest option	Extra parts are as wrong as missing parts.
Changing the shape	If the shape stays the same along a row, an option with a different shape is wrong (Eng 2019 option 3).
Counting parts wrongly	Count outlines and circles one by one; a single circle is not a double circle.
Mixing a row rule and a column rule	Decide once whether the matrix works along rows or columns.
Treating decoration as the rule	Details that differ in a way that follows no rule (such as the lines in Eng 2022 rows 1 and 2) can be ignored if another feature gives a rule.

Scope

Related PYQs / topics	Where
Figure series with a question mark at the end of a row (Eng 2019 Q36; Maths 2024 Q75, Q96; Science 2024 Q79, Q98; SS 2023 Q7, Q26)	Rule Detection / figure series (earlier topic)
Paper folding and cutting (Eng 2019 Q47, Q48, Q49)	Paper Folding (next)

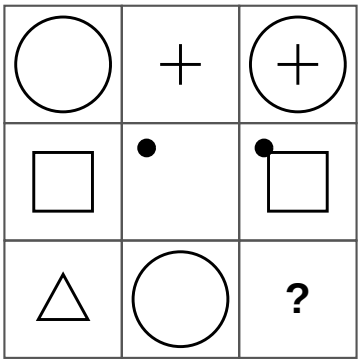
CHAPTER 5

Practice Set

6 original questions; answers on the next page (every option set verified by program)

A. Row rules: overlay and common-removal

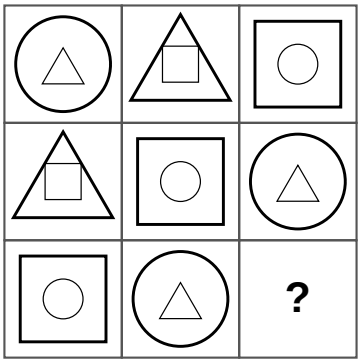
Q1. Which figure completes the matrix?





A B C D

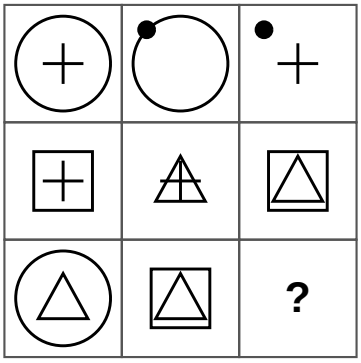
Q2. Which figure completes the matrix?





A B C D

Q3. Which figure completes the matrix?

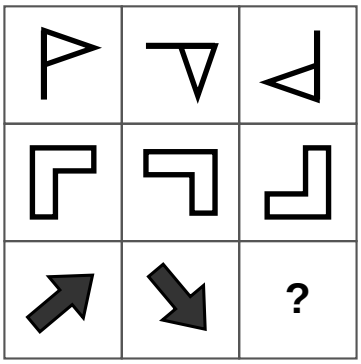




A B C D

B. Rotation and counting

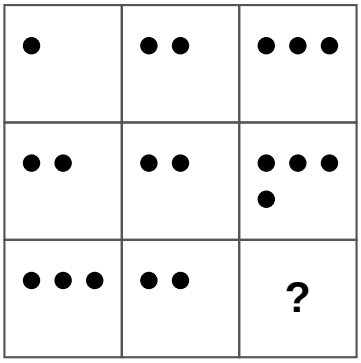
Q4. Which figure completes the matrix?





A B C D

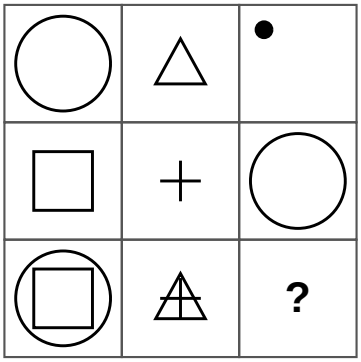
Q5. Which figure completes the matrix?

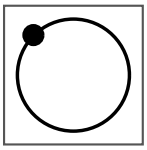




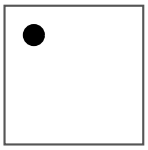
A B C D

Q6. Which figure completes the matrix? (Look at the columns.)

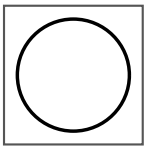




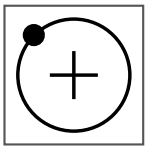
A



B



C



D

Answer key

Q	Answer	Reason
1	C	In every row the third figure is the first and second drawn on top of each other. Row 3: triangle + circle. A and B each show only one part; D adds a dot that is not in either figure.
2	A	Outer shapes cycle circle, triangle, square along each row and column, so every shape appears once in each row and column: row 3 needs a triangle outside. The inner shape is always the next one in the cycle (circle, triangle, square), so the inner shape is a square.
3	D	In each row, the part common to the first two figures disappears and the other parts make up the third. Row 3: the triangle is common, so circle and square remain. A keeps the common triangle; B and C are the first or a mixed figure.
4	B	In every row the figure turns a further 90° clockwise from one column to the next (0°, 90°, 180°). The missing figure is the arrow turned 180°. The other options show the same arrow turned 0°, 90° or 270°.
5	C	In every row the number of dots in the third figure is the sum of the first two (1+2=3, 2+2=4). Row 3: 3+2=5 dots.
6	A	This matrix works down the columns: the bottom figure is the top and middle figures overlaid. Column 3: dot + circle. The rows follow no rule, which is why rows must be tested first and then columns.

CHAPTER 6

Quick Revision

One page for the last day

Rule	Test	PYQ
Progression	A layer or part is added at each step along the line	Eng 2019 Q53
Overlay	Third = first + second	–
Common removal	Common parts vanish	–
Cyclic / Latin	Each shape once per row and column; inner shape = next in the cycle	Eng 2022 Q47
Rotation	Constant turn from cell to cell	–
Counting	Dots, lines or sides add or change by a fixed step	–

PYQ RULE: Both PYQs: question mark in the bottom-right cell, answer from one row. Predict the answer before looking, then reject options with a wrong shape, a missing layer or an extra part.

ROUTINE: Exam-day routine: list the features; test the rows; test the columns; check every complete line; predict; eliminate. 40 seconds per question. Next in the syllabus order: Paper Folding.